

HackAthena 2.0 Project Selection Dossier

Objective Technical & Strategic Evaluation of Three Architectural Options for 'Detection and Prevention of AI-Based Frauds'

TEAM EXECUTIVE MEMORANDUM • CONFIDENTIAL STRATEGIC BRIEFING

PURPOSE OF THIS DOSSIER: This document has been prepared for the hackathon team members to impartially evaluate and select our final project topic for HackAthena 2.0 (Jyothi Engineering College, Oct 5–7, 2026). It presents three fully formulated, highly competitive options across both non-technical (plain English) and deep-technical dimensions. No single option is pre-selected; each represents a distinct architectural philosophy, skill balance, and demonstration experience. Review this document, evaluate your personal strengths, and let us vote on the final direction.

1. Event Profile, Official Theme & Adjudication Context

Before evaluating individual ideas, the team must align on the binding parameters of the competition, sourced directly from the official HackAthena 2.0 portal and its registration code:

- **Host Institution & Format:** 48-Hour non-stop offline hackathon organized by the Computer Engineering Students Association (CESA) at Jyothi Engineering College (Autonomous), Cheruthuruthy, Thrissur, Kerala. Scheduled for October 5–7, 2026.
- **Binding Central Theme:** "Detection and Prevention of AI based Frauds" — Reimagine how fraud committed using AI, cloned voices, deepfake videos, synthetic faces, fabricated documents, and other AI-generated forgeries, is detected, prevented, and stopped across identities, transactions, and real-world systems.
- **Official Example Pillars / Tracks:** (1) Voice & Video Deepfakes, (2) Transaction Fraud, (3) Document Integrity, (4) Synthetic Identity & KYC, (5) Open Fraud Track.
- **Selection Round Mandates (Deadline: Sept 27):** Registration is performed by the Team Leader via Google OAuth. The submission strictly mandates a Google Drive Folder Link (set to 'Anyone with link can view') containing: (a) Project Abstract (PDF), and (b) Official Presentation (PPT) strictly adhering to the 5-slide official template downloaded from the portal.
- **Roster Rules:** Strict team size of 2 to 4 members. Registration fee of ₹1,200 per shortlisted team includes food and on-campus accommodation. Total prize pool: ₹90,000.

Host Jury Adjudication Realities at Jyothi Engineering College

Historical evaluation patterns from past HackAthena editions (e.g., L.I.O.N, LearnDrive.AI) demonstrate that the jury strongly favors: (1) Systems-level depth and technical authenticity, (2) 100% deterministic local execution running offline without dependency on external cloud APIs (e.g., OpenAI/Claude rate limits or institutional Wi-Fi dropouts), and (3) Visually inspectable causality (inputs clearly transform into verifiable mathematical outputs). Thin LLM wrappers and static mock dashboards are severely penalized.

2. Architectural Option 1: Pivoted ShadowGram

Behavioral Graph Detection of Coordinated AI-Agent Swarms & Botnets

ARCHITECTURAL THESIS

CORE PHILOSOPHY: Shift the detection paradigm from individual account inspection ('Is this account fake?') to relational graph intelligence ('Do these apparently independent accounts exhibit programmatic coordination indicating an AI agent swarm?').

2.1 The Concept in Plain English (The Non-Technical Story)

In 2026, malicious actors no longer deploy crude, repetitive spam bots. Instead, they use autonomous AI agents (powered by tools like Browser-Use, AutoGPT, and local LLMs) to create networks of synthetic accounts. Because an LLM writes unique text, picks different profile photos, and simulates conversational variety for each account, traditional security filters that look at single accounts are completely blind—each account looks like a normal, independent human.

However, because all these accounts are being commanded by an underlying automated agent script, their hidden behavior gives them away: they navigate web routes in identical mathematical sequences, their typing pauses follow programmatic delays rather than human physical anatomy, and their posting rhythms synchronize in micro-temporal bursts. ShadowGram instruments the web application to collect these subtle behavioral breadcrumbs, connects the accounts into a relationship web, and flags the coordinated AI swarm before it can manipulate opinions, execute wash trades, or harvest promotional bonuses. Its centerpiece is the 'Why are these accounts linked?' card, which gives compliance officers a plain-English breakdown of the evidence.

2.2 Technical Deep Dive & System Architecture

ShadowGram consists of a 4-tier pipeline designed for high-concurrency client telemetry ingestion and offline graph analytics:

1. Client Instrumentation SDK: A lightweight, unobtrusive JavaScript event collector embedded into client-side input forms and routing layers. It tracks: (a) Flight Time (FT) and Dwell Time (DT) variance across keystrokes, (b) Normalized pointer trajectory curvature and jerk, (c) Finite State Machine (FSM) navigation route transitions, and (d) Client canvas/WebGL rendering hashes.

2. Feature Engine & Embedding Worker: Normalizes client telemetry into continuous multi-variate feature vectors x in R^D . User-submitted text fields are asynchronously converted into dense 384-dimensional semantic vectors using sentence-transformers/all-MiniLM-L6-v2, enabling semantic cosine distance calculation across paraphrased AI text.

3. Multi-Signal Composite Graph Engine: Computes pairwise similarity across all active accounts using a multi-dimensional weighted formulation:

$$S_{\text{comp}}(u, v) = w_{\text{nav}} * S_{\text{nav}} + w_{\text{temp}} * S_{\text{temp}} + w_{\text{type}} * S_{\text{type}} + w_{\text{sem}} * S_{\text{sem}} + w_{\text{env}} * S_{\text{env}}$$

An undirected weighted edge is dynamically inserted into a NetworkX graph whenever $S_{\text{comp}}(u, v) \geq 0.78$ across at least three independent categories.

4. Community Detection & Explainability Layer: Executes Louvain modularity clustering to extract connected subgraphs representing coordinated syndicates. For every detected community, the system auto-generates a forensic attribution card itemizing the variance differentials (e.g., 'Typing Flight-Time Covariance: 0.93; Inter-Arrival Ingress Delta < 1.4s; Semantic Similarity: 0.89').

2.3 HackAthena Alignment, Tech Stack & Demo Walkthrough

Target Track / Pillar

Track 04: Synthetic Identity & KYC / Track 05: Open Fraud Track

Core Technology Stack	FastAPI (Python 3.14/uv), NetworkX, Sentence-Transformers, Next.js / React-Force-Graph, SQLite
Hardware / Offline Reqs	100% Offline CPU execution. Under 200MB RAM footprint. Sub-80ms embedding inference.
Synthetic Data Benchmark	Automated generator harness simulating N=100 accounts (70 organic Gaussian, 15 botnet, 10 farm, 5 outliers)
Live 3-Minute Pitch Script	0:00-0:45: Explain the AI agent swarm blindspot. 0:45-1:45: Launch 3 headless browser agent sessions; watch the real-time graph dynamically spin up and link them. 1:45-2:30: Open the 'Why' card to inspect the plain-English evidence trail. 2:30-3:00: Trigger automated cluster quarantine and display saved reach metrics.
Required Team Skillsets	1 Full-Stack / Graph UI Engineer (React/Cytoscape), 1 Backend Engineer (FastAPI/WebSockets), 1 ML/Data Engineer (NetworkX/Telemetry), 1 Presenter/Strategist

2.4 Objective Pros, Cons & Strategic Trade-Offs

- **Primary Strengths (Pros):**
 - + Unmatched visual presentation value: An interactive force-directed graph lighting up live creates an immediate psychological hook for judges.
 - + Explainability centerpiece: Directly addresses the black-box opacity critique that judges frequently raise against AI security tools.
 - + Highly resilient tech stack: Classical ML, graph theory, and local transformers run reliably without any GPU or cloud network dependencies.
- **Primary Limitations & Risks (Cons):**
 - Indirect AI Fraud Connection: Does not directly inspect AI-generated media (images, video, audio); detects behavioral coordination of the accounts operating them.
 - The Simulation Critique: Skeptical evaluators may point out that telemetry is captured inside an instrumented demo environment rather than a production consumer platform.

3. Architectural Option 2: DeepDoc / GhostKYC

Multimodal AI-Generated Document & Face Forensic Engine (Pure Document & Identity Forensics)

ARCHITECTURAL THESIS

CORE PHILOSOPHY: Direct forensic deconstruction. Every generative AI model leaves imperceptible mathematical and spectral footprints. Expose the diffusion frequency anomalies and inpainting compression differentials to prove forgery at the pixel and signal level.

3.1 The Concept in Plain English (The Non-Technical Story)

India's financial sector is currently experiencing an unprecedented wave of AI-enabled identity fraud. Criminals targeting fintech platforms (like Zerodha, Groww, and Paytm) or instant micro-loan apps take genuine photos of Aadhaar cards or PAN cards and use AI inpainting tools to alter the name, date of birth, and identity number while perfectly matching the official card's fonts, colors, and background textures. To complete the mandatory digital KYC video or photo check, they use generative AI (such as Midjourney or Flux) to generate photorealistic human faces that look 100% human but belong to nobody.

DeepDoc acts like a digital forensic laboratory built into the onboarding pipeline. Instead of just reading text with regular OCR, it conducts a multi-spectral medical exam on the document: it exposes the microscopic grid patterns left by AI image generators, detects the hidden compression seams where text was digitally painted over, and scans the secure QR code on the Aadhaar card to cross-check if the cryptographically signed government data matches the printed text. When an AI forgery is submitted, DeepDoc lights up the exact forged words in red bounding boxes and generates an auditable forensic certificate in under 1.5 seconds.

3.2 Technical Deep Dive & System Architecture

DeepDoc operates a 4-stage deterministic forensic pipeline executing entirely on CPU:

1. 2D Fast Fourier Transform (FFT) Spectral Analysis: Generative diffusion models and deep convolutional upsamplers introduce subtle, periodic high-frequency grid artifacts during the reverse-diffusion denoise process. The system converts the submitted face crop into the frequency domain via 2D FFT:

$$F(u, v) = \sum_x \sum_y f(x, y) * e^{-j * 2 * \pi * (ux/M + vy/N)}$$

By computing the azimuthal radial average of the magnitude spectrum $|F(u, v)|$, DeepDoc calculates the High-Frequency Energy Ratio (HFER) in the outer spectrum ($r > 0.65 r_{max}$). Synthetic diffusion faces show sharp, abnormal periodic energy spikes compared to organic camera sensors.

2. Error Level Analysis (ELA) & Tamper Localization: Digital inpainting alters the local error level potential across different compression macroblocks. DeepDoc resaves the document at a calibrated JPEG quality (90%), calculates the absolute pixel-wise difference $|I_{orig} - I_{resaved}|$, and applies dynamic thresholding. Areas modified by generative inpainting exhibit markedly higher compression differentials, yielding precise bounding boxes around forged text.

3. Cryptographic Cross-Modal Checksum Verification: Scans the embedded high-density QR code on Aadhaar/PAN cards using pyzbar/OpenCV. It unpacks the digitally signed XML/byte payload, computes the SHA-256 hash of the demographic data, and compares it character-by-character against the visual text extracted by Tesseract/easyOCR. Any mismatch between the cryptographic payload and the visual card text flags instant forgery.

4. Biometric Embedding Cross-Match: Utilizes a lightweight local ONNX facial embedding model to verify identity continuity between the ID photo crop and the live selfie frame, ensuring no synthetic proxy attack has occurred.

3.3 HackAthena Alignment, Tech Stack & Demo Walkthrough

Target Track / Pillar	Track 04: Synthetic Identity & KYC / Track 03: Document Integrity (Direct Bullseye)
Core Technology Stack	Python 3.14, OpenCV (cv2), SciPy (scipy.fft), PIL, PyTesseract / easyOCR, Pyzbar, FastAPI, Next.js UI
Hardware / Offline Reqs	100% Offline execution on standard laptop CPU. Zero cloud calls. Total analysis time < 1.2s per document.
Demonstration Benchmark	Curated test set of 20 genuine Aadhaar/PAN cards vs 20 AI-inpainted cards & 20 Midjourney/Flux synthetic selfies
Live 3-Minute Pitch Script	0:00-0:45: The Aadhaar/PAN AI forgery crisis in Indian banking. 0:45-1:30: Upload a genuine Aadhaar (system turns GREEN in 1s, verified QR). 1:30-2:15: Upload an AI-inpainted card (screen flashes RED, highlights forged name in bounding box, displays ELA heatmap). 2:15-2:45: Upload a synthetic selfie (displays 2D FFT spectral graph with diffusion artifact spike). 2:45-3:00: Download official PDF forensic audit dossier.
Required Team Skillsets	1 Computer Vision / DSP Engineer (OpenCV, FFT, ELA), 1 OCR/Parsing Engineer (Tesseract/QR), 1 Full-Stack Engineer (FastAPI/Next.js), 1 Presenter/Domain Specialist

3.4 Objective Pros, Cons & Strategic Trade-Offs

• **Primary Strengths (Pros):**

- + 100% Direct Hit on Live Theme: Directly attacks AI-generated faces and fabricated documents, perfectly aligning with Pillars 3 & 4.
- + Sensational, Incontrovertible Live Pitch: Judges physically watch a fake ID get exposed with exact pixel heatmaps and mathematical Fourier graphs.
- + Visceral Local Relevance: Indian academic and industry judges deeply relate to Aadhaar/PAN fraud and digital banking security in Kerala.

• **Primary Limitations & Risks (Cons):**

- Isolated Node-Level Scope: Inspects documents one at a time; does not understand or visualize coordinated multi-account bot syndicates.
- OCR Variance: Low-resolution or poorly lit mobile camera uploads can occasionally degrade OCR character extraction accuracy.

4. Architectural Option 3: GHOSTNET / AuraKYC

The Multimodal Forensic Sensors + Relational Syndicate Graph Fusion

ARCHITECTURAL THESIS

CORE PHILOSOPHY: Complete defensive spectrum. Single-point forensic analysis catches the forged artifact; relational graph intelligence exposes the entire criminal syndicate operating behind the scenes.

4.1 The Concept in Plain English (The Non-Technical Story)

Organized financial crime syndicates do not generate a single fake identity and stop there. They run automated pipelines that fabricate batches of 30 to 50 synthetic identities at once—using generative AI to create dozens of distinct faces and forged documents—to systematically loot instant loan applications or set up money mule rings. Catching one fake document is useful, but while you are banning that one account, the other 49 accounts have already cashed out.

GHOSTNET combines the forensic precision of Option 2 with the network intelligence of Option 1 into an all-in-one defense platform. At the front door, it uses digital forensics (FFT spectral analysis, ELA inpainting detection, and QR verification) to inspect the applicant's ID card and selfie. Simultaneously, it feeds all onboarding metadata—such as the digital fingerprints of the generative AI model, identical browser hardware hashes, and synchronized application submission bursts—into a real-time relationship graph. The moment a single account fails forensic verification, GHOSTNET lights up the entire hidden syndicate on an interactive web map, allowing a bank to freeze all 50 fraudulent accounts in a single click.

4.2 Technical Deep Dive & System Architecture

GHOSTNET implements a dual-tier, multi-modal architecture orchestrated by an asynchronous event bus:

- Tier 1: Forensic Sensor Array (Node Level):** When an account registers, the forensic worker executes 2D FFT spectral analysis on the selfie and ELA tamper localization on the ID card in parallel. It outputs a scalar AI Fraud Probability P_{node} in $[0, 1]$ along with an extracted latent artifact vector ϕ_{artifact} .
- Tier 2: Relational Syndicate Clustering (Graph Level):** The graph worker ingests: (a) ϕ_{artifact} (generator model noise signatures), (b) client environment hashes (Canvas/WebGL fingerprints), (c) inter-arrival registration intervals Δt , and (d) document layout structural similarity. It constructs a multi-layer NetworkX graph where edge weights represent cross-account coordination confidence.
- Tier 3: Graph Topology & Syndicate Isolation:** Applies Louvain community detection to segment the graph into independent clusters. If any node within a cluster exhibits $P_{\text{node}} > 0.80$, the entire connected subgraph is elevated to high-risk syndicate status.
- Tier 4: Unified Dual-Pane Compliance Dashboard:** Built in Next.js. The left pane renders an interactive force-directed graph displaying all onboarded accounts, highlighting flagged syndicates in pulsating crimson. Clicking any node immediately populates the right pane with the deep forensic dossier (ELA tamper heatmaps, 2D Fourier graphs, and plain-English coordination causality).

4.3 HackAthena Alignment, Tech Stack & Demo Walkthrough

Target Track / Pillar	Track 04: Synthetic Identity & KYC / Track 03: Document Integrity / Track 02: Transaction Fraud
Core Technology Stack	FastAPI backend, OpenCV & SciPy (FFT/ELA), NetworkX (Graph analytics), Next.js with React-Force-Graph
Hardware / Offline Reqs	100% Offline execution on standard laptop CPU. Ingestion latency < 1.4s per account; graph re-clustering < 120ms.

Demonstration Benchmark	Synthetic banking population of N=100 accounts containing 2 coordinated criminal syndicates (15 synthetic accounts each)
Live 3-Minute Pitch Script	0:00-0:40: The Dual Challenge: AI forgeries at the node + automated syndicates across the network. 0:40-1:30: Individual node demo: Upload an AI-inpainted Aadhaar, displaying real-time Fourier spikes and tamper heatmaps. 1:30-2:20: Zoom out to the NetworkX graph: show how catching this single node unmasks 14 other coordinated accounts sharing generator seeds and canvas hashes. 2:20-3:00: Execute 'Quarantine Syndicate' button, demonstrating zero-leakage containment.
Required Team Skillsets	1 Computer Vision Specialist (OpenCV/FFT), 1 Graph/ML Specialist (NetworkX/Clustering), 1 Full-Stack Backend Engineer (FastAPI/Async), 1 Frontend & UI Specialist (Next.js/Canvas)

4.4 Objective Pros, Cons & Strategic Trade-Offs

- **Primary Strengths (Pros):**
 - + Unmatched Architectural Scope: Solves the blind spots of both Option 1 (adds concrete AI forensic analysis) and Option 2 (adds multi-account network clustering).
 - + Exceptional Storytelling Arc: The pitch transitions smoothly from a micro forensic inspection to a macro syndicate takedown.
 - + Cross-Track Versatility: Strong alignment across three different official pillars (Synthetic Identity, Document Integrity, and Transaction Fraud).
- **Primary Limitations & Risks (Cons):**
 - Higher 48-Hour Implementation Complexity: Requires integrating both a computer vision pipeline and a graph analytics pipeline into a unified UI.
 - Tight Team Coordination Required: Demands clear API contracts between the CV engineer, graph engineer, and frontend developer early on.

5. Side-by-Side Comprehensive Comparison Matrix

The following matrix contrasts the three architectural candidates across twelve critical hackathon dimensions to help the team evaluate technical feasibility, demo reliability, and judging dynamics impartially:

Evaluation Dimension	Option 1: Pivoted ShadowGram	Option 2: DeepDoc / GhostKYC	Option 3: GHOSTNET / AuraKYC
Core Defense Target	AI Agent Swarms & Botnets	AI-Inpainted IDs & Synthetic Faces	Synthetic Identity Rings & Syndicates
Primary Official Pillar	Track 04 (Identity) & 05 (Open)	Track 04 (Identity) & 03 (Documents)	Tracks 04, 03, and 02 (Unified)
Core Algorithmic Domain	Graph Theory & NLP Embeddings	Computer Vision & Signal Processing (DSP)	Multimodal DSP + Graph Theory
Key Mathematical Tools	Louvain Modularity, Cosine Distance	2D Fast Fourier Transform, ELA Calculus	2D FFT, ELA, Graph Adjacency Matrices
Live Demonstration Hook	Real-time Force-Directed Graph	Pixel Heatmaps & FFT Spectral Graphs	Dual-Pane: Forensic Heatmaps + Network Graph
48h Development Complexity	Moderate (Standard Web + NetworkX)	Moderate (CV Pipeline + Dashboard)	High (CV + Graph + Unified Frontend)
Resistance to Jury Critique	Moderate (Simulated client telemetry)	Exceptional (Concrete physical artifacts)	Exceptional (Inspects artifacts + network)
Local Offline Reliability	100% Deterministic (Zero Cloud APIs)	100% Deterministic (Zero Cloud APIs)	100% Deterministic (Zero Cloud APIs)
Ideal Team Skill Profile	Web Full-Stack + Data Science	Computer Vision / Image Processing	Balanced Team (CV + Graph + Full-Stack)
Primary Live Demo Risk	Scripted bot jitter during live demo	OCR lighting variance on sample cards	Pipeline latency during dual-pane render
Pitch Narrative Clarity	High ('Catching automated swarms')	Immediate ('Catching forged Aadhaar')	Very High ('Catching the whole syndicate')

6. Decision-Making Framework & Team Voting Guide

To select the best project objectively, our team should not vote based on what sounds coolest in theory, but rather on what maximizes our competitive edge given our team's specific distribution of skills and the 48-hour build constraints. Review the four voting criteria below:

Scenario-Based Voting Rules

VOTE PROFILE: OPTION 1

- VOTE FOR OPTION 1 (PIVOTED SHADOWGRAM) IF:
 - Our team consists primarily of full-stack web developers (React/Next.js/FastAPI) and data analysts.
 - We want a polished, highly interactive, and beautiful frontend UI without having to write low-level computer vision or image processing algorithms.
 - We feel confident scripting simulated browser agents and explaining graph clustering theory to judges.

VOTE PROFILE: OPTION 2

- VOTE FOR OPTION 2 (DEEPDOG / GHOSTKYC) IF:
 - At least one or two team members are enthusiastic about computer vision, signal processing (OpenCV, Fourier transforms), and OCR.
 - We want the most direct, visceral, and unarguable live demonstration possible (holding up real vs. forged Aadhaar/PAN cards and synthetic faces).
 - We want to avoid graph databases, clustering mathematics, or browser telemetry entirely and focus on building a laser-focused forensic tool.

VOTE PROFILE: OPTION 3

- VOTE FOR OPTION 3 (GHOSTNET / AURAKYC) IF:
 - Our team has balanced, complementary skills (e.g., one person strong in OpenCV/DSP, one strong in Graph/Backend, and one or two in Full-Stack/UI).
 - We want to build an ambitious, flagship-level hackathon project that covers the full defensive spectrum (individual forensic analysis + syndicate network intelligence).
 - We are disciplined enough to divide and conquer the architecture early with clear API schemas so integration proceeds smoothly.

Immediate Next Steps Once the Team Votes

1. Finalize Team Roster: Confirm all 2 to 4 team members (Full Names, College Register Numbers, Emails, Phone Numbers, and B.Tech Courses).
2. Populate the Official 5-Slide Presentation: Transfer the chosen idea's problem statement, approach, implementation architecture, and impact into the official template PPTX.
3. Export the 2-Page Project Abstract: Generate the clean PDF document summarizing the project.
4. Google Drive Submission Setup: Place both files into a dedicated Google Drive folder, verify sharing is set to 'Anyone with the link can view', and submit on the HackAthena portal before the September 27 deadline.